

TAN YU HOE

Singapore

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WORK EXPERIENCE

HTX (Home Team Science and Technology Agency)

Singapore

AI Engineer and Project Management Intern, CNPMC

Apr 2026 – Jul 2026

- Detention and Custodial Solutions, Correction and Narcotics Programme Management Centre
- Built Qwen3-VL photo-moderation pipeline for SPS ePhoto Submission on NGINE, a sovereign air-gapped AI infrastructure, using zero-shot visual grounding to flag prohibited content with category labels, bounding-box evidence, reviewer-facing reasoning traces, and evaluation reports.

GovTech DSAID (Data Science and AI Division)

Singapore

AI Engineer Intern, Video Analytics Team

Apr 2022 – Jun 2023

- Main engineer for photo relevance study in municipal feedback cases for MSO (Municipal Services Office). Proposed and developed an Image Relevance model using zero-shot image classification and embedding similarity methods. Achieved 97% accuracy without model training.
- Evaluated the performance of zero-shot visual-language models (CLIP and OWL-ViT) as general-purpose computer vision capabilities, for visual recognition use cases with limited labelled data.
- Implemented interactive web interfaces and backend APIs for zero-shot object detection, using Gradio, FastAPI, PyTorch, Docker, and HuggingFace transformer SDK.
- Conducted a comparison study between MMDetection framework and YOLO Darknet framework for pedestrian pathway object detection.
- Presented multiple knowledge-sharing sessions on open-vocabulary models, visual-language pretraining (VLP) models, and self-supervised learning (SSL).

EDUCATION

National University of Singapore

Singapore

Bachelor of Science (Honours) in Business Analytics

Aug 2025 – Present

- Specialization in Machine Learning Analytics
- Minor in Statistics

College of Alice and Peter Tan

Singapore

University Town College Programme

Aug 2025 – May 2027

- UTC1409 The Pursuit of Happiness
- UTW1001T How Rich Should Anyone Be? The Ethics of Wealth Inequality
- CAPT Renew Engager (HEB Ashram Halfway House)
- CAPTuring Dreams (=DREAMS (Singapore) Limited)
- House Committee - Member

Singapore Polytechnic

Singapore

Diploma in Applied AI and Analytics

Apr 2020 – May 2023

- **Honors:** 3.81 / 4.0 GPA
- **Courses:** Statistic, Python for Data Analytics, AI & Machine Learning, Deep Learning
- **CCA Grade:** Gold with Honours
- **Distinctions:** ST0503 Back-End Web Development, ST1511 AI and Machine Learning, ST1504 Deep Learning, ST3004 Infocomm Professional Seminar

LEADERSHIP AND ACTIVITIES

SP Artificial Intelligence Student Chapter

Singapore

Operations Head

Dec 2020 – Mar 2022

- Led a project team that developed a headline-based fake news detection application for showcase in Singapore Polytechnic's Open House and Engineering Show 2022

- Orchestrated a Kaggle competition on a curated dataset with real-world applications, attracting 70+ participants and fostering skill development in machine learning and data analysis techniques
- Curated teaching materials and taught concepts on Data Science workflows and Machine Learning concepts for two data-science-related workshops

HONORS AND AWARDS

Trace | Ship It Hack'26 2nd Place (powered by Replit) Singapore

Issued by *Ship It Hacks* Jun 2026

- Won 2nd place for Trace, an agentic CCTV indexing and search engine that helps security teams upload surveillance footage, automatically index frames, and search for people, vehicles, objects, or events using natural language.
- Built the Replit Agent-provisioned React/Express/PostgreSQL/GCS pipeline with ffmpeg frame extraction and GPT-5-mini vision analysis.
- Returned timestamped frame previews and relevance-ranked results, compressing hours of manual CCTV review into a faster investigative search workflow.

NUS Hack4Good 2026 - Second Place Singapore

Issued by *National University of Singapore* Jan 2026

- Top 2 out of 57 teams
- Built Granted, an automated grant discovery engine to help Non-Profit Organisations (NPOs) find and act on funding opportunities from the Singapore Government (OurSG Grants).
- Built a Firebase Cloud Functions backend with 24 production-ready functions for NPO auth, grant discovery, saved grants, profile management, CORS, and error handling.
- Implemented OpenAI-embedding grant matching with scheduled CRON jobs, HTTP APIs, Firestore-triggered updates, weekly match emails, and OurSGGrants ingestion.

NUS LifeHack 2023 - First Place + sgID Award Singapore

Issued by *NUS Computing Club* Jun 2023

- Won first place and sgID Award (bonus) by Open Government Products
- Developed AquaWise, an AI-powered sustainable shopping companion focused on mindful consumption and integrated with e-commerce platforms such as Shopee.

WPH Digital Hackathon 2022 - Second Place Award Singapore

Issued by *WPH Digital* Apr 2022

- Developed an innovative solution to streamline CV screening, addressing the inefficiencies in identifying top talent. Collaborated with a team of tech enthusiasts to create a transformative tool that simplifies the hiring process, showcasing problem-solving skills and a commitment to reshaping the future of the job market.

LICENSES AND CERTIFICATIONS

Google Data Analytics ID: 5D46PZTLVV9Q

Issued by *Coursera* Apr 2023

Google IT Automation with Python ID: 878DAB3LU5FY

Issued by *Coursera* Jan 2023

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Native Speaker), Chinese (Native Speaker)
- **Programming:** Python, SQL, Javascript, Java, R
- **Tools:** Git, GitHub, Docker, GCP, Visual Studio Code, Ubuntu/Linux
- **Cloud & Backend:** Firebase Cloud Functions, Firestore, RESTful APIs, Serverless Architecture
- **BI & Data Analytics:** Tableau, Jupyter Notebook, Numpy, Pandas, Matplotlib/Seaborn, Plotly, Statistics, RStudio
- **Machine Learning:** Scikit-Learn, TensorFlow, PyTorch, PyTorch Lightning, HuggingFace, OpenAI API, MMDetection